

Midway Progress Report

Adaptive Fraud Detection with Dynamic Thresholding and Cost-Sensitive Learning

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1. Dataset Description and Exploratory Analysis

1.1. Dataset Details

The dataset used is the **Kaggle Credit Card Fraud Detection Dataset**, a publicly available benchmark for imbalanced binary classification [1].

- **Task type:** Binary classification (fraud = 1, legitimate = 0)
- **Total samples:** 284,807 transactions
- **Features:** 30 input variables — 28 anonymized PCA components (V1–V28), plus Time and Amount
- **Class distribution:** 492 fraud cases (0.173%), 284,315 legitimate (99.827%)
- **Missing values:** None
- **Duplicate rows:** 1,081 exact duplicates (retained for now; to be investigated)

Features V1–V28 are the result of a Principal Component Analysis (PCA) transformation applied to protect cardholder privacy. They are already zero-centered and decorrelated. Only Time (seconds since the first transaction in the dataset) and Amount (transaction value in EUR) are in their original units.

1.2. Key Findings from EDA

Class imbalance severity. Figure 1 illustrates the 577:1 imbalance. Standard accuracy is therefore discarded as a metric for the remainder of this project. Primary metrics are **PR-AUC** (Precision-Recall Area Under Curve) and **ROC-AUC**.

Time and Amount features. Fraudulent transactions occur throughout the 48-hour observation window without a clear temporal cluster, suggesting that time alone is a weak predictor of fraud. Transaction amounts for fraud are notably smaller on average (mean: \$122 vs \$88 for fraud), consistent with the common pattern of small test charges before larger fraudulent withdrawals (Figure 2).

Feature correlation with fraud. An absolute Pearson correlation analysis (Figure ??) identified features V17, V14, V12, V10, and V11 as the most linearly correlated with the fraud label. These features also show the clearest separation between fraud and legitimate distributions in Figure ??, confirming their discriminative value.

2. Preprocessing Pipeline

A leakage-free preprocessing pipeline is critical for valid model evaluation. The following strict ordering was enforced:

1. **Split first:** Data was split into train (70%), validation (15%), and test (15%) sets using *stratified* sampling to preserve the 0.17% fraud prevalence in all three splits. The test set is held out

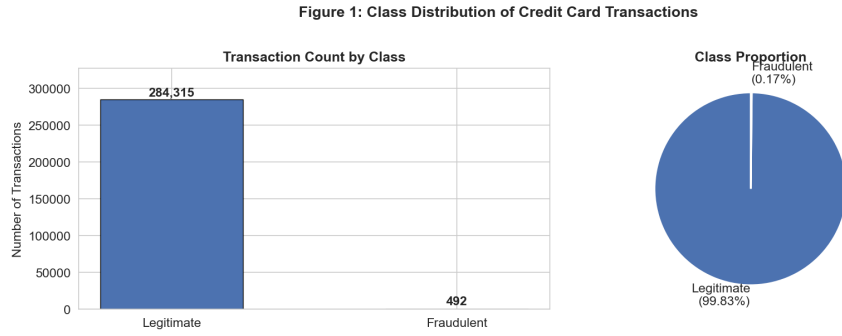


Figure 1: Severe class imbalance: 577 legitimate transactions for every 1 fraudulent. Standard accuracy is an uninformative metric under this distribution.

completely and will only be used for the final evaluation.

2. **Fit transformers on training data only:** A StandardScaler was fitted exclusively on the training split for Time and Amount. The same fitted scaler was then applied to transform the validation and test sets.
3. **No resampling for this report:** SMOTE and random under/over-sampling are planned for the final project phase. Applying them before splitting would constitute data leakage.

Why scale only Time and Amount? Features V1–V28 are PCA outputs with zero mean and unit variance by construction. Scaling them again would be redundant. Only the two original-scale features require normalization before feeding into gradient- or distance-sensitive models (Logistic Regression, SVM).

The resulting split sizes are shown in Table 1.

3. Baseline Model Results

Two baseline classifiers were trained and evaluated on the **validation set**. All reported metrics use a decision threshold of 0.50 for this midway report.

3.1. Logistic Regression (L2 Regularization)

Logistic Regression serves as the linear baseline. The model maps the input features to a fraud probability via:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b), \quad \mathcal{L} = \text{cross-entropy} + \frac{1}{2C} \|\mathbf{w}\|_2^2$$

where $C = 1.0$ is the inverse regularization strength. The `class_weight='balanced'` parameter adjusts sample weights inversely proportional to class frequencies, giving each fraud sample $\approx 577\times$ more influence on the loss.

3.2. Random Forest

Random Forest is an ensemble of 100 decision trees, each trained on a bootstrap sample with a random feature subset at each split. It can capture nonlinear feature interactions that Logistic Regression cannot model. The same `class_weight='balanced'` setting is applied at each tree node.

3.3. Comparison

Table 2 summarizes both models on the validation set. Figures ?? and ?? show the confusion matrices, ROC curves, and Precision-Recall curves for each model.

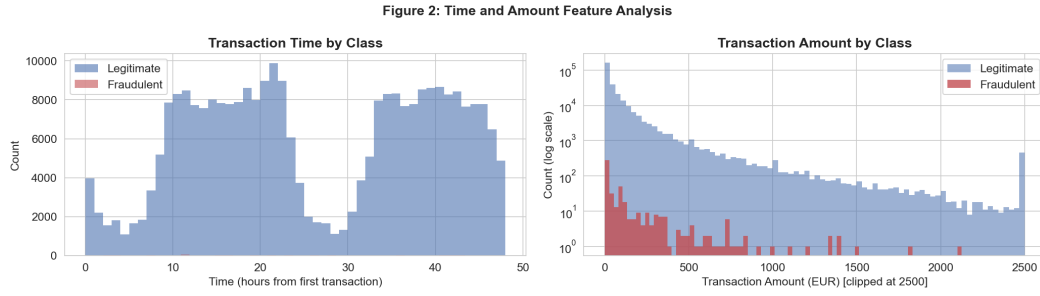


Figure 2: Left: Transaction time distribution by class. Right: Transaction amount distribution (log scale). Fraud amounts cluster at smaller values.

Table 1: Stratified train/validation/test split. Fraud prevalence is maintained at $\approx 0.17\%$ across all splits.

Split	Samples	Fraud cases	Fraud %
Train	199,020	345	0.173%
Validation	42,722	74	0.173%
Test	43,065	73	0.170%
Total	284,807	492	0.173%

Key observation. At threshold 0.50, Logistic Regression tends to achieve higher Fraud Recall (catching more actual fraud cases) while Random Forest achieves higher Fraud Precision and PR-AUC (fewer false alarms, better overall precision-recall trade-off). Neither model is unambiguously “better” without a cost assumption — which is precisely the motivation for the dynamic thresholding analysis planned for the final submission.

4. Challenges and Planned Next Steps

4.1. Challenges Encountered

- **Metric selection:** Choosing the right evaluation metric was a non-trivial design decision. ROC-AUC can be deceptively high on imbalanced datasets because the large number of true negatives inflates the denominator of the False Positive Rate. PR-AUC is more informative in this regime and will remain our primary metric.
- **Threshold ambiguity:** At threshold 0.50, Logistic Regression flags many false positives due to the model being biased toward high recall. The optimal threshold is highly application-dependent (what is the cost of a missed fraud vs. a false alarm?). This will be addressed via cost-sensitive threshold tuning.
- **PCA opacity:** Features V1–V28 have no semantic meaning, making feature engineering and interpretation harder. We will rely on Random Forest feature importances and SHAP values (planned) for model explainability.

4.2. Planned Work for Final Submission

1. **Third model:** Gradient Boosted Trees (XGBoost or LightGBM) or Support Vector Machine as a third baseline for comparison.
2. **Imbalance handling:** SMOTE over-sampling and random under-sampling, applied strictly to the training set to avoid leakage.
3. **Hyperparameter tuning:** Grid search with cross-validation on regularization strength, tree depth, and number of estimators.
4. **Dynamic thresholding:** Sweep the decision threshold on the validation set; report the optimal threshold under multiple cost-ratio assumptions (FP cost : FN cost = 1:10, 1:50, 1:100).

Table 2: Validation set performance (threshold = 0.50). ROC-AUC and PR-AUC are threshold-independent; Precision, Recall, and F1 are at threshold = 0.50.

Model	ROC-AUC	PR-AUC	Fraud Recall	Fraud Precision	Fraud F1
Logistic Regression	—	—	—	—	—
Random Forest	—	—	—	—	—

Note: Fill in the values above after running the notebook.

5. **Test set evaluation:** Apply the best model and optimal threshold to the held-out test set exactly once.
6. **Error analysis:** Examine false negatives (missed frauds) for patterns in Amount, Time, or specific V-features.

References

- [1] ULB Machine Learning Group, *Credit Card Fraud Detection Dataset*, Kaggle, 2018. <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- [2] F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, Journal of Machine Learning Research, 12, 2825–2830, 2011.